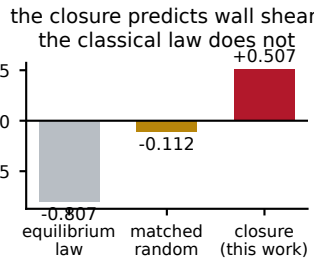
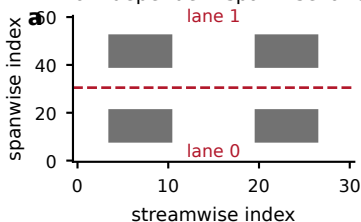
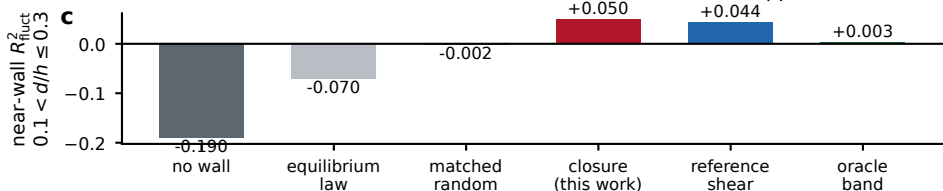


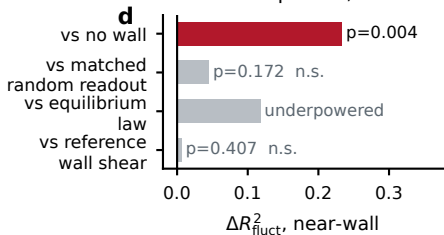
four cubic roughness elements,
two independent spanwise lanes



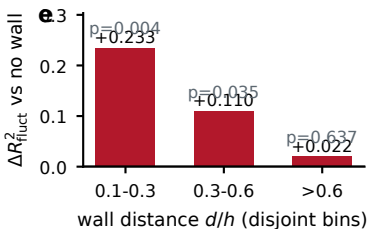
what each wall channel is worth, on cells that were neither supplied nor read



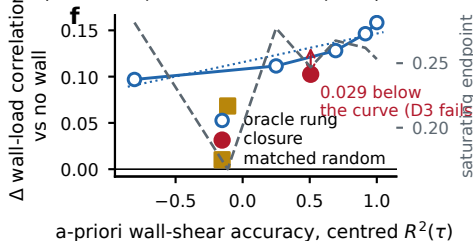
registered attribution:
one clause passes, one fails



the effect is wall-concentrated



wall-flux accuracy controls the wall load
 $\rho = +0.93$, $p = 0.0034$; flat endpoint $\rho = -0.25$



NOT a wall force: no direction, no face a
no form drag. Wrong-instant is LOWER than

